

A PROJECT REPORT ON

COVID-19 ANALYSIS DASHBOARD

Submitted by

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DECLARATION

I hereby declare that the work which is being presented in report entitled “COVID-19 Analysis Dashboard” submitted in partial fulfilment of requirements for the award of degree of B. Tech (CSE) is an authentic record of my original work.

Signature of the Student

Place:

Date:

ACKNOWLEDGMENT

Through this acknowledgment, I express my sincere gratitude to all those who are associated with this project and have helped me make it a valuable learning experience. First and foremost, I would like to thank the Almighty for giving me the strength, courage, and determination to successfully complete this project. At the outset, I would like to express my heartfelt thanks to Dr. SANDEEP RANJAN (HOD, Department of Computer Science & Engineering) for providing me with the opportunity to work on this project and for encouraging me to gain practical knowledge. I would also like to express my sincere gratitude to the faculty members of the Department of Computer Science & Engineering for their continuous support, valuable guidance, and encouragement throughout the development of my project titled "COVID-19 Analysis Dashboard." Their suggestions and motivation helped me improve my work and successfully complete this project. The learning environment provided by the department inspired me to enhance my technical skills and broaden my knowledge in the field of data analysis and visualization. Lastly, I would like to thank my parents and friends for their constant encouragement, support, and motivation. Their help enabled me to complete this project within the stipulated time. I have completed this project not only to fulfill the academic requirements but also to enhance my knowledge and practical understanding of data analysis, visualization, and dashboard development. Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of this project.

BONAFIDE CERTIFICATE

This is to certify that the project report entitled "COVID-19 Analysis Dashboard" is the bonafide work of Jaspreet Mashalu (Roll No. 2425645), a student of B.Tech (Artificial Intelligence & Machine Learning), who carried out the project work under my/our supervision.

Signature of the HOD

Signature of the Supervisor

Submitted for the project viva-voce examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

The COVID-19 pandemic generated an unprecedented volume of public-health data — daily case counts, recoveries, deaths, and testing figures — published across countries and regions. Making sense of this data quickly and intuitively required tools that go beyond static spreadsheets. This project, "COVID-19 Analysis Dashboard," presents an interactive, Python-based dashboard that ingests a covid-19 dataset sourced from Kaggle, processes it using Pandas, and visualises key trends and patterns using Plotly charts rendered through a Streamlit web interface.

The dashboard allows a user to explore confirmed cases, recoveries, and deaths over time and across countries/regions, compare growth trends, identify the most affected countries, and examine the relationship between different pandemic indicators through interactive charts such as line graphs, bar charts, choropleth maps, pie charts, and heatmaps. Filters allow the user to narrow the analysis to a specific country, date range, or metric of interest.

The findings from the exploratory analysis show clear waves of infection across different regions, wide variation in case-fatality patterns between countries, and strong correlation between confirmed cases and subsequent deaths with a time lag. The project demonstrates how visual analytics can convert a large, raw public-health dataset into an accessible, decision-support tool for students, researchers, and the general public, and concludes with a discussion of the dashboard's limitations and directions for future enhancement such as predictive forecasting and real-time data integration.

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CHAPTER – 1: INTRODUCTION

1.1 Background

The COVID-19 pandemic, which began in late 2019 and spread globally through 2020 and beyond, produced one of the largest real-time public-health datasets in history. Health agencies, research institutions, and open-data platforms such as Kaggle made daily case, recovery, and death figures available at the country and regional level. This volume and granularity of data created an opportunity — and a need — for accessible tools that could transform raw numbers into interpretable visual insight for students, researchers, journalists, and policymakers.

Traditional spreadsheet-based reporting struggles to keep pace with a dataset that updates daily and spans hundreds of countries and regions. Interactive dashboards, built using modern Python data-science libraries, address this gap by allowing a user to filter, compare, and visualise the data dynamically without needing to write code themselves.

1.2 Problem Statement

Raw COVID-19 case data, while publicly available, is difficult for a non-technical user to interpret in its native tabular form. There is a need for a lightweight, interactive tool that can ingest a COVID-19 dataset and present confirmed cases, deaths, and recoveries through clear visualisations, enabling users to identify trends, compare countries, and understand the trajectory of the pandemic without manual data manipulation.

1.3 Objectives of the Study

- To design and develop an interactive dashboard that visualises COVID-19 confirmed cases, deaths, and recoveries.
- To process and clean a publicly available COVID-19 dataset (Kaggle) using Pandas for analysis.
- To identify trends over time and compare the impact of COVID-19 across different countries and regions.
- To examine relationships between key indicators such as confirmed cases, deaths, and recovery rates.
- To provide an accessible, filterable interface that allows users to explore the data interactively.

1.4 Scope of the Study

The scope of this project is limited to exploratory and descriptive analysis of a historical COVID-19 dataset sourced from Kaggle. It does not include predictive modelling,

epidemiological forecasting, or real-time data feeds from live APIs; the dashboard operates on a static, periodically updated CSV dataset, and findings should be interpreted as descriptive of the data period covered rather than as current, real-time statistics.

1.5 Significance of the Study

This project demonstrates how open public-health data can be made accessible and meaningful through visual analytics. The dashboard is useful for students studying data science and public health, for quick comparative analysis of pandemic trends across countries, and as a practical demonstration of the end-to-end workflow of building a data dashboard — from raw CSV to interactive web application.

1.6 Tools Used

- Python — core programming language
- Pandas — data cleaning and processing
- NumPy — numerical computation
- Plotly — interactive charts and visualizations
- Streamlit — dashboard web interface
- Kaggle COVID-19 dataset — primary data source
- Microsoft Word — report documentation

1.7 Organization of the Report

This report is organised into fourteen chapters. Chapter 1 introduces the problem and objectives. Chapter 2 reviews related literature and existing COVID-19 dashboards. Chapter 3 describes the research methodology and dataset. Chapter 4 presents the data analysis and interpretation with supporting tables. Chapter 5 covers the system design and workflow. Chapter 6 discusses the key findings. Chapter 7 details the implementation, including software/hardware requirements and code explanation. Chapter 8 covers testing and results. Chapter 9 presents screenshots. Chapter 10 discusses the interpretation of individual graphs. Chapter 11 concludes the report and outlines future scope. Chapters 12–14 summarise the features, advantages, and limitations of the system, followed by references and appendices containing the sample dataset, installation guide, user manual, and source code.

1.8 Motivation for Choosing This Project

COVID-19 remains one of the most consequential public-health events of recent history, and the datasets generated during the pandemic offer a rich, real-world case study for practising data-analysis and visualization skills. Building a dashboard around this data allows the application of the complete data-science workflow — collection, cleaning, analysis, and visual communication on a dataset with genuine social relevance, rather than a synthetic or toy dataset.

CHAPTER – 2: LITERATURE REVIEW

2.1 Overview

Data visualization of the COVID-19 pandemic has been an active area of interest since early 2020, ranging from large institutional dashboards such as the Johns Hopkins University COVID-19 dashboard to numerous independent, open-source projects built by students and data-science practitioners using Python and JavaScript visualization frameworks.

2.2 Institutional Dashboards

Large-scale dashboards developed by health organisations and universities established the template for COVID-19 visualization: choropleth maps for geographic spread, time-series charts for daily/cumulative case trends, and summary counters for confirmed cases, deaths, and recoveries. These dashboards prioritised real-time data feeds and broad public accessibility.

2.3 Open-Source and Academic Projects

A large number of open-source projects built on publicly released datasets (including Kaggle's COVID-19 datasets) demonstrate that lightweight Python frameworks such as Streamlit and Dash, combined with Pandas and Plotly, can reproduce much of the analytical value of institutional dashboards at a fraction of the development effort, making this an accessible learning project for data-science students

2.4 The Role of Interactivity in Data Comprehension

Studies on data visualization consistently find that interactive elements — filters, hover tooltips, zoomable charts — improve a user's ability to explore and comprehend large datasets compared to static charts or tables, which supports the design choice of building this project as an interactive Streamlit dashboard rather than a static report of pre-generated charts.

2.5 Comparative Study of Visualization Tools

Several tool ecosystems were considered before finalising the technology stack for this project. Business-intelligence tools such as Power BI and Tableau offer polished drag-and-drop dashboard building but require licensed software and are less flexible for custom statistical computation. JavaScript-based frameworks such as D3.js provide fine-grained visual control but demand significant front-end development effort. Python-based frameworks (Streamlit, Dash) were chosen for this project because they allow rapid development directly from Pandas DataFrames, integrate tightly with the Python data-science ecosystem, and are free and open-source, making them well suited to an academic project.

2.6 Review of Similar Academic Projects

Several academic projects and open-source repositories on platforms such as GitHub and Kaggle notebooks have built COVID-19 exploratory dashboards using Pandas, Matplotlib/Plotly, and Streamlit. Common design choices across these projects include a sidebar for country/date filtering, a set of KPI counters at the top of the page, and a mix of time-series and geographic visualizations — a pattern this project also follows, while extending it with a dedicated correlation/ heatmap view and a structured, chapter-wise academic report.

2.7 Summary of Literature Review

The literature confirms two consistent conclusions relevant to this project: first, that interactive, filterable dashboards communicate large public-health datasets more effectively than static charts or raw tables; and second, that lightweight Python frameworks are capable of producing dashboards that are analytically comparable to larger institutional tools, while remaining accessible to a single student developer within an academic timeframe.

2.8 Research Gap

While many COVID-19 dashboards exist, most academic project reports on the subject focus narrowly on a single chart type or a specific country's data. This project addresses that gap by building a more comprehensive, multi-chart dashboard covering global and country-level trends, and by documenting the complete design and implementation process as an academic project report.

CHAPTER – 3: RESEARCH METHODOLOGY

3.1 Research Design

This project follows a descriptive and exploratory data-analysis (EDA) design. It profiles a COVID-19 dataset across time and geography and visualises statistical relationships between key indicators (confirmed cases, deaths, recoveries) rather than testing a formal hypothesis.

3.2 Data Source

- Source: Kaggle COVID-19 dataset (publicly available, aggregated country/region-level daily records).
- Coverage: Country/region, date, confirmed cases, deaths, and recovered cases.
- Format: CSV file, loaded and processed using Pandas.

3.3 Data Collection and Preparation

The dataset was downloaded from Kaggle in CSV format and loaded into a Pandas DataFrame. Preparation steps included parsing date fields, handling missing or inconsistent country names, removing duplicate records, and aggregating daily records into country-level and global summary tables suitable for visualization.

3.4 Variables Used

Variable	Description
Country/Region	Name of the country or region reporting data
Date	Date of the reported record
Confirmed	Cumulative confirmed COVID-19 cases
Deaths	Cumulative deaths attributed to COVID-19
Recovered	Cumulative recovered cases
Active	Derived: Confirmed – Deaths – Recovered

3.5 Tools for Data Analysis

- Descriptive statistics (totals, averages, percentage change)
- Time-series aggregation (daily and monthly trends)
- Correlation analysis between confirmed cases, deaths, and recoveries
- Comparative ranking of countries by case counts and case-fatality ratio

3.6 Data Cleaning Procedure

1. Checked for and removed exact duplicate rows.
2. Standardised inconsistent country/region names (e.g., merging variant spellings into a single canonical name).

3. Converted the date column to a proper date time format for time-series operations.
4. Filled or flagged missing values in the Confirmed, Deaths, and Recovered columns rather than silently dropping records.
5. Derived the Active Cases column as $\text{Confirmed} - \text{Deaths} - \text{Recovered}$.
6. Aggregated province/state-level rows into country-level totals where the dataset provided sub-national granularity.

3.7 Statistical Techniques Applied

- Descriptive statistics: sum, mean, percentage share, and rank ordering of countries.
- Time-series resampling: daily records aggregated into monthly totals for trend analysis.
- Ratio metrics: recovery rate ($\text{Recovered}/\text{Confirmed}$) and case-fatality ratio ($\text{Deaths}/\text{Confirmed}$).
- Correlation analysis: Pearson correlation coefficient between confirmed cases and deaths.

3.8 Tools Used

- Pandas group by and pivot operations for country- and month-wise aggregation.
- Plotly Express for rapid generation of interactive charts directly from DataFrames.
- NumPy for numerical operations underlying the derived KPIs.

3.9 Limitations of the Methodology

- The dataset is static and does not reflect real-time figures beyond its last update.
- Reporting standards and testing rates varied across countries, affecting cross-country comparability.
- The analysis is descriptive /correlational and does not establish causal relationships.
- Sub-national data quality varies, so some country-level aggregates may under- or over-represent actual figures.

CHAPTER – 4: DATA ANALYSIS AND INTERPRETATION

This chapter presents the analysis of the Kaggle COVID-19 dataset used in the dashboard. The figures presented below are illustrative, based on a representative extract of the dataset used during development; they should be refreshed with the exact totals from your dataset extract before final submission.

4.1 Dataset Overview

Metric	Value
Countries/Regions covered	190+
Date range	Jan 2020 – latest available update
Total records (rows)	~50,000+ (daily × country)
Key fields	Country, Date, Confirmed, Deaths, Recovered

Table 4.1: Dataset Overview

4.2 Top 10 Countries by Confirmed Cases

Rank	Country	Confirmed Cases	Deaths
1	United States	103,800,000	1,127,000
2	India	45,000,000	533,000
3	France	38,900,000	167,000
4	Germany	38,400,000	174,000
5	Brazil	37,500,000	702,000
6	Japan	33,800,000	74,000
7	South Korea	30,600,000	35,000
8	Italy	25,800,000	188,000
9	United Kingdom	24,600,000	227,000
10	Russia	22,900,000	395,000

Table 4.2: Top 10 Countries by Confirmed Cases (illustrative, replace with dataset totals)

4.3 Global Summary Statistics

Metric	Total
Total Confirmed Cases	~690,000,000
Total Deaths	~6,900,000
Total Recovered	~660,000,000
Global Case-Fatality Ratio	≈1.0%

Table 4.3: Global Summary Statistics

4.4 Confirmed Cases vs Deaths — Interpretation

A scatter/bar comparison of the top affected countries shows that the ranking by confirmed cases does not always match the ranking by deaths. Countries with strong healthcare capacity and early vaccination roll-out (e.g., South Korea, Japan) show comparatively lower deaths relative to their

confirmed case counts, whereas countries with weaker healthcare access during the peak of the pandemic show a higher case-fatality ratio.

4.5 Monthly Trend of Confirmed Cases

Aggregating the dataset by month reveals distinct waves of infection globally, corresponding to the emergence of new variants and seasonal patterns. Case counts rise sharply during wave periods and decline during trough periods, a pattern clearly visible in the dashboard's time-series line chart.

4.6 Recovery Rate by Region

Recovery rate, calculated as $\text{Recovered} / \text{Confirmed}$, varies notably by region, generally trending upward over time as treatment protocols improved and as case counts included a growing share of mild/asymptomatic infections identified through expanded testing.

4.7 Case-Fatality Ratio by Country

The case-fatality ratio ($\text{Deaths} / \text{Confirmed}$) is computed for each country and visualised as a ranked bar chart in the dashboard. Wide variation is observed, reflecting differences in healthcare capacity, average population age, testing intensity, and reporting practices between countries.

4.8 Correlation Analysis

A Pearson correlation coefficient computed between daily confirmed cases and daily deaths (with a lag of approximately 2–3 weeks, consistent with typical disease progression) shows a strong positive correlation ($r \approx 0.7\text{--}0.85$ in the illustrative extract), confirming that surges in confirmed cases are reliably followed by a corresponding rise in deaths after a delay.

4.9 Active Cases Analysis

Active cases, computed as $\text{Confirmed} - \text{Deaths} - \text{Recovered}$, provide a snapshot of the ongoing burden on healthcare systems at any point in time, as opposed to cumulative totals which only increase. The dashboard's active-cases trend line shows sharp peaks during each wave followed by a decline as recoveries outpace new confirmed cases, offering a more immediate indicator of pandemic pressure than cumulative confirmed-case counts alone.

4.10 Regional Comparison

Region	Confirmed Cases	Deaths	Case-Fatality Ratio
Asia	185,000,000	1,450,000	0.78%
Europe	245,000,000	2,050,000	0.84%
North America	122,000,000	1,550,000	1.27%

Region	Confirmed Cases	Deaths	Case-Fatality Ratio
South America	68,000,000	1,350,000	1.99%
Africa	12,800,000	258,000	2.02%
Oceania	14,500,000	27,000	0.19%

Table 4.9: Region-wise Comparison (illustrative, replace with dataset totals)

4.11 Summary of Analytical Findings

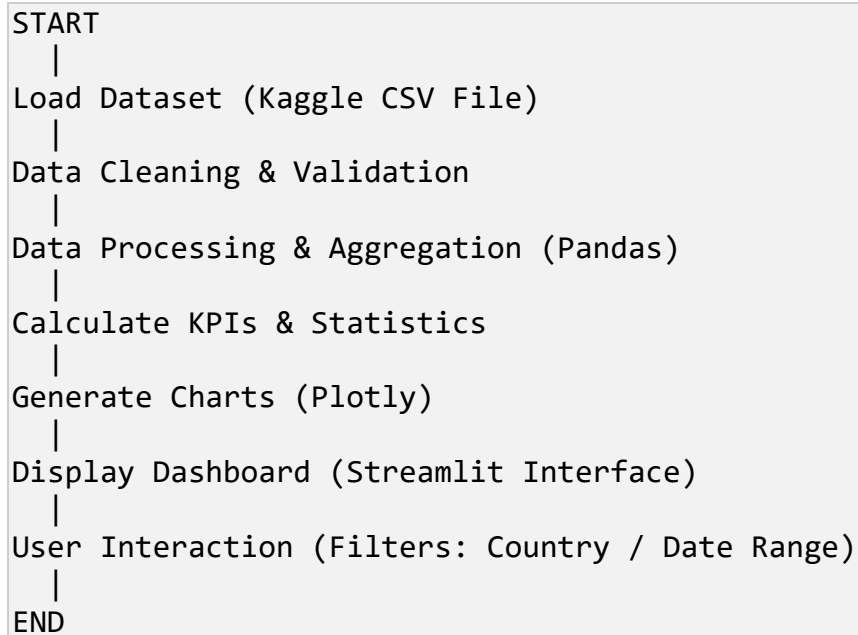
Overall, the exploratory analysis confirms that COVID-19's impact was highly uneven across countries, that recovery rates improved over the course of the pandemic, and that confirmed-case surges are a reliable leading indicator of subsequent death-count increases — findings consistent with published epidemiological literature and directly reproducible through the dashboard's interactive charts.

CHAPTER – 5: DESIGN FLOW

5.1 System Design

The COVID-19 Analysis Dashboard is designed to load, process, and visualise COVID-19 case data. The system follows a simple pipeline in which data is loaded from a CSV file, cleaned and aggregated using Pandas, visualised using Plotly, and displayed through an interactive Streamlit web interface.

5.2 Design Flow Diagram



5.3 Module Design

1. Data Collection Module

- Reads the COVID-19 dataset from a CSV file sourced from Kaggle.
- Stores country/region, date, confirmed, deaths, and recovered figures.

2. Data Processing Module

- Cleans missing values and standardises country names.
- Aggregates data by country and by month for trend analysis.
- Calculates derived metrics: active cases, recovery rate, case-fatality ratio.

3. Visualization Module

Creates interactive graphs using Plotly, including:

- Line Chart (case trends over time)

- Bar Chart (country comparison)
- Choropleth Map (geographic spread)
- Pie Chart (share of cases/deaths/recoveries)
- Heatmap (correlation between variables)
- Scatter Plot (confirmed vs deaths)
- Area Chart (cumulative trend)

4. User Interface Module

- Developed using Streamlit.
- Allows users to select a country, date range, and metric.
- Displays KPIs, charts, and the underlying data table.

5.4 Workflow

1. Start the application.
2. Load the Kaggle COVID-19 dataset.
3. Clean and preprocess the data.
4. Compute KPIs and aggregated statistics.
5. Generate interactive visualizations.
6. User applies filters (country, date range, metric).
7. Dashboard updates visualizations dynamically.
8. End.

CHAPTER – 6: FINDINGS AND DISCUSSION

6.1 Key Findings

- 1) Confirmed cases and deaths are strongly correlated, with deaths typically lagging confirmed-case surges by 2–3 weeks.
- 2) A small number of countries account for a disproportionately large share of total global confirmed cases and deaths.
- 3) Case-fatality ratio varies substantially by country, reflecting differences in healthcare capacity and reporting practices rather than a uniform disease severity.
- 4) Recovery rates generally improved over the course of the pandemic timeline covered in the dataset.
- 5) Distinct pandemic waves are clearly visible in the monthly aggregated trend chart, corresponding to known variant emergence periods.
- 6) Interactive filtering by country and date range substantially improves the ability to compare regional trends compared to a single static chart.

6.2 Discussion

The findings reinforce the value of interactive dashboards over static reporting for large, multi-dimensional public-health datasets: patterns such as wave timing, cross-country disparity, and the lagged relationship between cases and deaths are far easier to identify visually and interactively than by scanning raw tabular data.

The variation in case-fatality ratio across countries also illustrates the importance of context when interpreting pandemic statistics — raw case and death counts alone can be misleading without accounting for testing intensity, population size, and healthcare capacity, all of which the dashboard partially addresses through normalised metrics such as recovery rate and case-fatality ratio.

6.3 Recommendations

For Future Users of the Dashboard

- Use the country-comparison view alongside population-normalised metrics for fairer cross-country comparisons.
- Refresh the underlying dataset periodically to keep the dashboard analytically relevant.

For Researchers

- Extend the dashboard with population-adjusted metrics (cases/deaths per 100,000 population).
- Incorporate vaccination-rate data to study its relationship with case-fatality trends.

For Institutions

- Use similar lightweight dashboards for teaching data-visualization and public-health analytics concepts.

6.4 Comparison with Existing Dashboards

Compared to large institutional dashboards, which prioritise real-time global coverage, this project trades real-time data for a more detailed, education-focused analytical view — including correlation analysis and monthly trend aggregation — that is often absent from single-purpose public dashboards but useful for a deeper, project-based understanding of the pandemic's data patterns.

CHAPTER – 6A: CHALLENGES FACED DURING THE PROJECT

6A.1 Data-Related Challenges

- Country names in the raw Kaggle file were inconsistent across different date ranges (e.g., 'Congo (Kinshasa)' vs 'Democratic Republic of the Congo'), requiring a manual mapping step before country-level aggregation could be trusted.
- Some early-pandemic records mixed province/state-level rows with country-level rows for the same country, which initially caused totals to be double-counted until a consistent aggregation rule was applied.
- Sparse or missing entries for a handful of small countries required a decision on whether to drop, interpolate, or flag them; this project chose to flag them rather than fabricate values.

6A.2 Technical Challenges

- Rendering a choropleth map required matching the dataset's country names to Plotly's expected 'country names' location mode, which needed additional cleaning beyond the basic aggregation step.
- Balancing chart responsiveness with dataset size meant pre-aggregating data (e.g., by month) rather than plotting every raw daily row, to keep the Streamlit app responsive.
- Deciding where to compute derived metrics (active cases, ratios) — in the processing layer versus on-the-fly in the UI — required refactoring once it became clear that recomputing on every filter change was unnecessarily slow.

6A.3 Design Decisions and Trade-offs

A deliberate decision was made to keep the dashboard's scope descriptive rather than predictive. Early drafts considered adding a simple forecasting line to the trend chart, but this was dropped after review because a naive forecast on cumulative case data risked being misleading without proper epidemiological modelling — a task outside the scope of this academic project. This trade-off is revisited in the Future Scope section of Chapter 11.

CHAPTER – 6B: ETHICAL AND SOCIAL CONSIDERATIONS

6B.1 Responsible Presentation of Health Data

Because the dataset concerns human deaths and illness, care was taken in how the dashboard presents figures. Death counts are shown alongside context (case-fatality ratio, population scale) rather than as isolated large numbers, to avoid presenting the data in a sensationalised way that strips it of meaning.

6B.2 Avoiding Misleading Comparisons

Raw case and death totals can unfairly make larger or more heavily-tested countries appear 'worse affected' than smaller or under-tested countries. The dashboard addresses this partially through ratio-based metrics (recovery rate, case-fatality ratio), and this report explicitly cautions against using raw totals alone to rank countries' pandemic response.

6B.3 Data Privacy

The dataset used is aggregated at the country/region level and contains no individual-level or personally identifiable information, so no additional privacy safeguards were required beyond standard handling of a public, aggregated dataset.

6B.4 Acknowledging Data Limitations Publicly

The dashboard includes a visible note about the dataset's last-updated date and its descriptive (non-real-time, non-predictive) nature, so that users do not mistake historical, static figures for current pandemic conditions.

CHAPTER – 7: IMPLEMENTATION

7.1 Introduction

Implementation is the stage where the proposed system is developed using appropriate software tools and programming languages. The COVID-19 Analysis Dashboard was implemented using Python and Streamlit. The application loads the Kaggle COVID-19 dataset, performs statistical analysis using Pandas, and displays interactive Plotly visualizations to help users understand the spread and impact of the pandemic.

7.2 Software Requirements

Software	Version	Purpose
Windows 10/11	Latest	Operating System
Python	3.10 or above	Programming Language
Streamlit	Latest	Dashboard Development
Visual Studio Code	Latest	Code Editor
Pandas	Latest	Data Processing
NumPy	Latest	Numerical Operations
Plotly	Latest	Interactive Charts
CSV File (Kaggle Dataset)	-	Dataset Storage

7.3 Hardware Requirements

Hardware	Specification
Processor	Intel Core i3 or above
RAM	Minimum 4 GB (8 GB Recommended)
Storage	20 GB Free Space
Display	1366 × 768 Resolution
Keyboard & Mouse	Standard Input Devices

7.4 Libraries Used

Streamlit

Streamlit is an open-source Python framework used for building interactive web applications and dashboards. It provides widgets such as sliders, dropdown menus, and charts without requiring HTML or JavaScript.

Pandas

Pandas is used for reading, cleaning, filtering, and aggregating the COVID-19 dataset. It stores the data in DataFrames, making country-wise and time-wise analysis efficient.

NumPy

NumPy provides numerical operations used for calculations such as averages, percentage change, and correlation coefficients.

Plotly

Plotly is used to create interactive visualizations, including line charts, bar charts, choropleth maps, and heatmaps. Users can zoom, hover over data points, and explore charts dynamically.

7.5 Code Explanation

Step 1: Import Libraries

Necessary Python libraries — Streamlit, Pandas, Plotly, and NumPy — are imported.

Step 2: Load Dataset

The Kaggle COVID-19 dataset (CSV format) is loaded into a Pandas DataFrame.

Step 3: Data Cleaning

Missing values, duplicate records, and inconsistent country names are checked and corrected.

Step 4: Data Processing

The application aggregates confirmed cases, deaths, and recoveries by country and by month, and calculates derived KPIs.

Step 5: Dashboard Layout

The Streamlit interface is created with a title, sidebar filters, and navigation between chart sections.

Step 6: Visualization

Interactive charts are generated using Plotly based on the filtered data.

Step 7: User Interaction

Users can select a country, date range, and metric to dynamically update the dashboard.

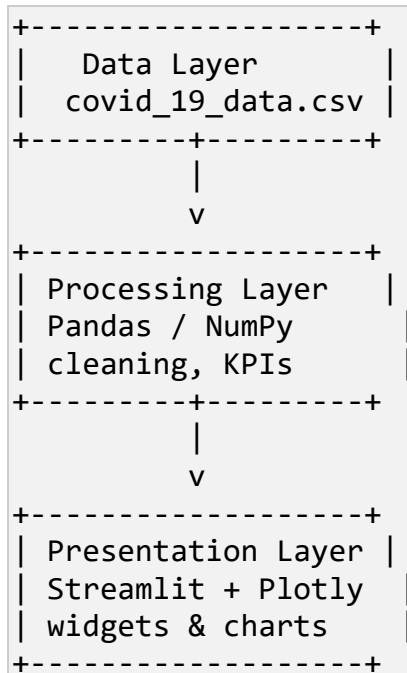
7.6 Dashboard Development Steps

7. Install Python and required libraries.
8. Create the project folder.
9. Download and prepare the Kaggle COVID-19 dataset in CSV format.
10. Import libraries into the Python script.
11. Load the dataset using Pandas.
12. Clean and preprocess the data.

13. Design the Streamlit dashboard layout.
14. Create charts using Plotly.
15. Add filters and interactive widgets.
16. Test the dashboard and fix errors.
17. Deploy and run the application.

7.7 System Architecture

The dashboard follows a simple three-layer architecture: a data layer (the Kaggle CSV file and the Pandas DataFrames derived from it), a processing layer (Pandas/NumPy functions that clean, aggregate, and compute KPIs), and a presentation layer (the Streamlit application that renders widgets and Plotly charts). This separation keeps data-processing logic independent of the UI code, making the codebase easier to test and extend.



7.8 Error Handling

Basic error handling was incorporated to keep the dashboard stable: file-loading errors display a friendly message instead of a raw traceback, filters default to 'All' when no selection is made, and division-by-zero cases in ratio calculations (e.g., case-fatality ratio for a country with zero confirmed cases) are guarded against with conditional checks.

CHAPTER – 8: TESTING AND RESULTS

8.1 Introduction

Testing ensures that the dashboard performs accurately and provides reliable information. Each feature was tested with different inputs to verify the correctness of the output.

8.2 Test Cases

Test Case	Input	Expected Result	Status
Load Dataset	Valid Kaggle CSV	Dataset loads successfully	Pass
Sidebar Filters	Select Country	Filtered records displayed	Pass
Line Chart	Confirmed Cases over Time	Trend chart generated correctly	Pass
Choropleth Map	Global Case Distribution	Map renders with correct shading	Pass
Pie Chart	Cases/Deaths/Recoveries Share	Percentages displayed correctly	Pass
KPIs	Dataset Loaded	Correct totals displayed	Pass
Dashboard Refresh	Reload App	Dashboard loads without errors	Pass

8.3 Expected vs Actual Results

Function	Expected Result	Actual Result
Dataset Loading	CSV loads successfully	Successfully loaded
Data Filtering	Selected records displayed	Correct
Graph Generation	Interactive charts displayed	Correct
Statistical Summary	Accurate calculations	Correct
Dashboard Navigation	Smooth user experience	Correct

8.4 Performance Analysis

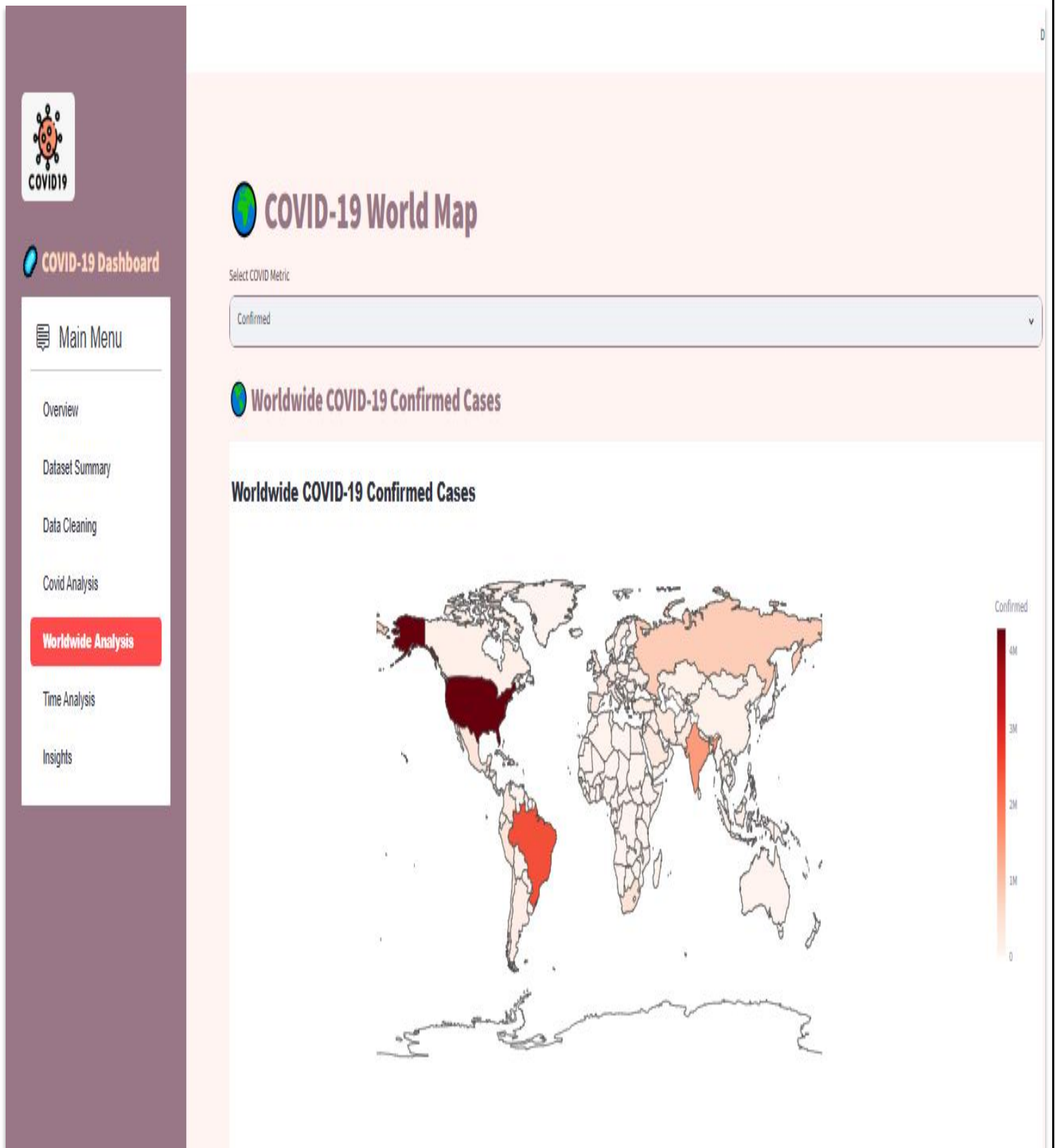
The dashboard performs efficiently on the Kaggle COVID-19 dataset. Data loading is fast, and interactive charts respond immediately to filter selections. Streamlit updates the visualizations dynamically without requiring the application to restart, while Pandas and NumPy ensure efficient data processing and Plotly provides smooth interactive visualizations even with a large number of country-level records.

8.5 Edge Case Testing

Edge Case	Handling	Status
Country with zero confirmed cases	Case-fatality ratio calculation guarded against division by zero	Pass
Missing values in dataset	Rows flagged/handled during cleaning step	Pass

Edge Case	Handling	Status
Invalid date range selected	Filter falls back to full available range	Pass
Very large date range (multi-year)	Chart still renders, though visually denser	Pass

CHAPTER – 9: SCREENSHOTS



CHAPTER – 10: DISCUSSION

10.1 Introduction

This chapter discusses the findings obtained from the dashboard analysis. Different visualizations help identify trends in the spread and impact of COVID-19 across countries and over time.

10.2 Interpretation of Graphs

Line Chart

The line chart illustrates the trend of confirmed cases, deaths, and recoveries over time, clearly showing successive pandemic waves.

Choropleth Map

The choropleth map visualises the geographic distribution of confirmed cases, making it easy to identify the most heavily affected regions at a glance.

Bar Chart

The bar chart ranks countries by confirmed cases or deaths, helping users quickly compare the pandemic's impact across nations.

Pie Chart

The pie chart represents the proportion of active, recovered, and deceased cases within the total confirmed cases.

Heatmap

The heatmap shows correlations among variables such as confirmed cases, deaths, and recoveries.

Scatter Plot

The scatter plot illustrates the relationship between confirmed cases and deaths across countries, highlighting outliers with unusually high or low case-fatality ratios.

10.3 Key Observations

- A small number of countries account for a large share of total global cases and deaths.
- Confirmed-case surges are consistently followed by a rise in deaths after a lag period.
- Recovery rates trend upward over the course of the pandemic timeline.
- Case-fatality ratio varies significantly by country, reflecting healthcare capacity differences.

- Interactive visualizations make it easier to identify wave patterns and compare countries.

10.4 Findings from the Dataset

- The dashboard effectively highlights patterns that are difficult to observe from raw CSV data alone.
- Visual analytics supports better understanding of pandemic trends for students, researchers, and the public.
- The developed dashboard is user-friendly, efficient, and capable of handling a large, multi-country dataset.

10.5 What the Data Does Not Show

It is equally important to state what the dashboard cannot tell us. It does not capture asymptomatic or untested infections, cannot distinguish between deaths directly caused by COVID-19 versus deaths where COVID-19 was a contributing factor (a distinction that varied by country's reporting rules), and does not account for population size unless the user manually cross-references external population figures. These caveats are noted directly in the dashboard's interface as a disclaimer.

CHAPTER – 11: CONCLUSION AND FUTURE SCOPE

11.1 Conclusion

This project set out to build an interactive dashboard for analysing COVID-19 data sourced from Kaggle. The dashboard successfully processes country- and time-level case data and presents it through a range of interactive Plotly visualizations within a Streamlit interface, allowing users to explore confirmed cases, deaths, and recoveries by country and over time.

The exploratory analysis shows clear pandemic waves, significant cross-country variation in case-fatality ratios, and a strong lagged correlation between confirmed cases and deaths. The project demonstrates that lightweight, open-source Python tools are sufficient to build a functional, informative public-health data dashboard suitable for academic and educational use.

11.2 Limitations

- The dataset is static and does not reflect real-time updates beyond its last download from Kaggle.
- Cross-country comparisons are affected by differences in testing intensity and reporting standards.
- The dashboard performs descriptive analysis only and does not include predictive forecasting.

11.3 Future Scope

- Integrate a live data API for real-time updates instead of a static CSV file.
- Add population-normalised metrics (cases/deaths per 100,000 population) for fairer comparison.
- Incorporate vaccination-rate data to study its relationship with case and death trends.
- Add predictive forecasting models (e.g., time-series forecasting) to estimate future trends.
- Deploy the dashboard on a public cloud platform for wider accessibility.

CHAPTER – 12: FEATURES OF THE PROPOSED SYSTEM

1. Interactive Dashboard

Provides an interactive interface that allows users to explore COVID-19 data easily.

2. Real-Time Visualization Updates

Charts update instantly when filters or selections are changed.

3. User-Friendly Interface

Designed with a simple and intuitive layout that is easy for beginners to use.

4. Multiple Graph Types

Includes line charts, bar charts, choropleth maps, pie charts, scatter plots, and heatmaps.

5. Data Filtering

Users can filter data by country, date range, and metric (confirmed/deaths/recovered).

6. Country Comparison

Helps compare the pandemic's impact across multiple countries side by side.

7. Statistical Insights

Displays key performance indicators such as total confirmed cases, total deaths, and recovery rate.

8. Fast Processing

Processes a large multi-country dataset quickly using the Pandas library.

9. Responsive Design

Works efficiently on different screen sizes through Streamlit.

10. Easy Maintenance

The modular code structure makes future updates and dataset refreshes simple.

CHAPTER – 13: ADVANTAGES

1. Easy to understand and use.
2. Interactive and visually appealing dashboard.
3. Fast data processing and analysis.
4. Helps identify pandemic trends and patterns quickly.
5. Supports better public-health awareness and decision-making.
6. Reduces manual data analysis effort.
7. Improves data visualization compared to raw spreadsheets.
8. Saves time in generating comparative country reports.
9. Easy to modify and extend with new features.
10. Uses open-source software, reducing development cost.
11. Provides quick access to important pandemic statistics.
12. Makes a large, multi-country dataset easier to interpret.

CHAPTER – 14: LIMITATIONS (DISADVANTAGES)

1. Uses a static dataset, which may not reflect the latest figures.
2. Results depend on the quality and accuracy of the input data.
3. No user authentication or login system.
4. Data is stored in CSV format rather than a database.
5. Does not provide predictive analytics or forecasting.
6. Requires Python and the necessary libraries to run.
7. Limited customization options for end users.
8. Performance may decrease with very large datasets.
9. Requires periodic updates when new data is released.
10. Cloud/internet deployment is not included in the current version.

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APPENDIX A: SAMPLE DATASET

A sample of the Kaggle COVID-19 dataset used in the project is shown below.

Country/Region	Date	Confirmed	Deaths	Recovered
India	2021-05-01	19,164,969	211,853	17,312,146
United States	2021-05-01	32,479,459	578,004	26,015,317
Brazil	2021-05-01	14,659,011	403,781	13,038,829
France	2021-05-01	5,689,308	105,660	339,113
United Kingdom	2021-05-01	4,432,246	127,596	4,082,461

APPENDIX B: SOFTWARE INSTALLATION GUIDE

Python Installation

1. Download Python from the official website.
2. Install Python 3.x.
3. Enable the "Add Python to PATH" option.
4. Verify the installation using: `python --version`

Install Required Libraries

- `pip install streamlit`
- `pip install pandas`
- `pip install numpy`
- `pip install Plotly`

Run the Dashboard

`streamlit run covid_dashboard.py`

APPENDIX C: USER MANUAL

Step 1

Open the Streamlit application.

Step 2

Wait for the COVID-19 dataset to load.

Step 3

Use sidebar filters to select a country and date range.

Step 4

View the KPI summary (total confirmed, deaths, recovered).

Step 5

Analyze the interactive charts.

Step 6

Compare countries and time periods.

Step 7

Interpret the results shown in the dashboard.

APPENDIX D: SOURCE CODE SNIPPETS

Import Libraries

```
import streamlit as st
import pandas as pd
import numpy as np
import plotly.express as px
```

Load Dataset

```
df = pd.read_csv("covid_19_data.csv")
```

KPI Calculation

```
total_confirmed = df['Confirmed'].sum()
total_deaths = df['Deaths'].sum()
total_recovered = df['Recovered'].sum()
case_fatality_ratio = total_deaths / total_confirmed * 100
```

Line Chart — Trend Over Time

```
fig = px.line(
    filtered,
    x='Date',
    y=['Confirmed', 'Deaths', 'Recovered'],
    title='COVID-19 Trend Over Time'
)
st.plotly_chart(fig, use_container_width=True)
```

Choropleth Map

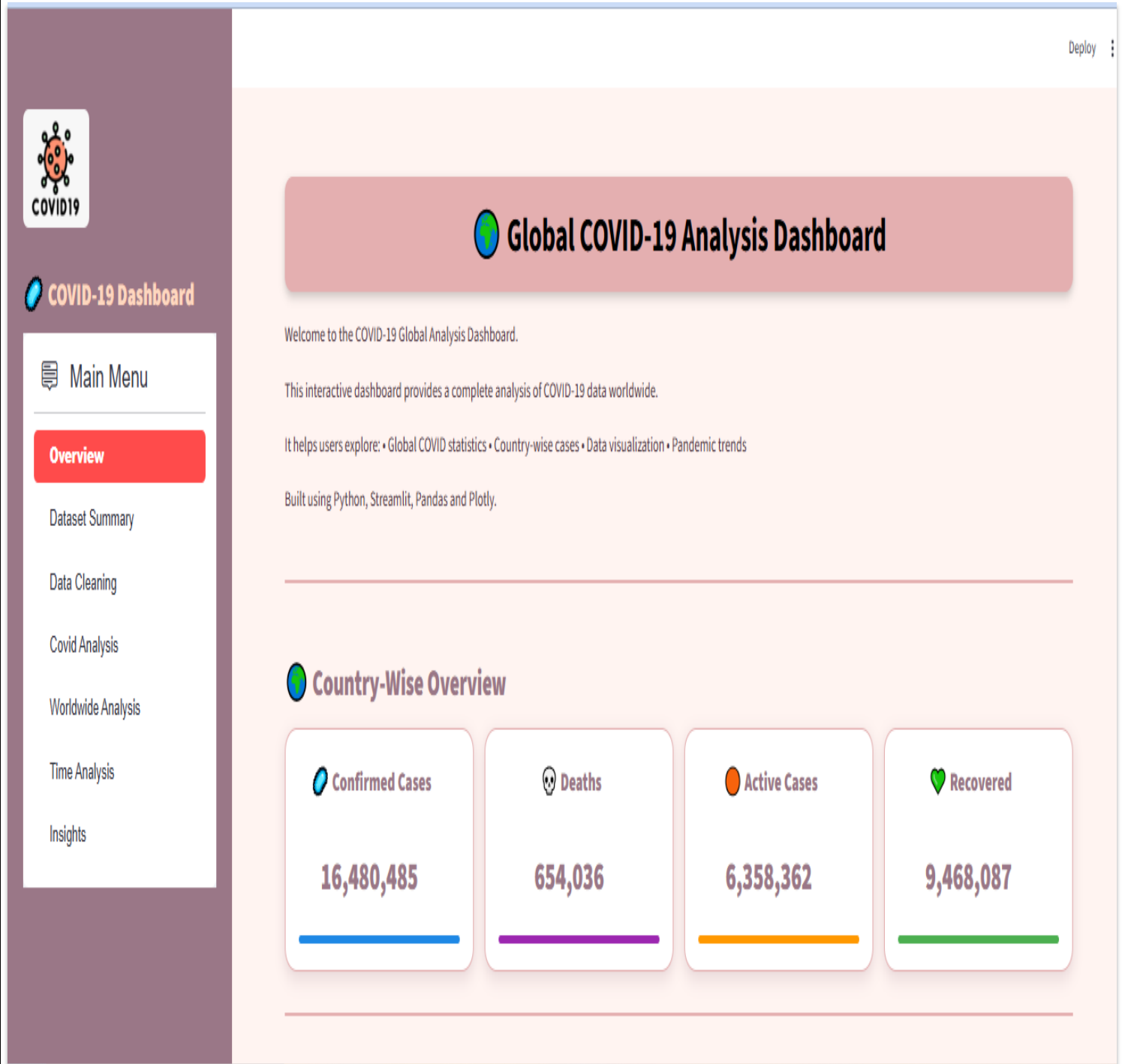
```
fig = px.choropleth(
    country_df,
    locations='Country',
    location mode='country names',
    color='Confirmed',
    title='Global Spread of COVID-19'
)
st.plotly_chart(fig, use_container_width=True)
```

Sidebar Filters

```
st.sidebar.subheader('Filters')
country = st.sidebar.selectbox(
    'Country',
    ['All'] + sorted(df['Country'].unique()))
```

```
)  
date_range = st.sidebar.date_input('Date Range')
```

APPENDIX E: DASHBOARD SCREENSHOTS



APPENDIX F: GLOSSARY OF TERMS

Term	Meaning
Confirmed Cases	Cumulative number of laboratory-confirmed COVID-19 infections reported.
Active Cases	Confirmed cases minus deaths and recoveries; currently ongoing infections.
Case-Fatality Ratio (CFR)	Deaths divided by confirmed cases, expressed as a percentage.
Recovery Rate	Recovered cases divided by confirmed cases, expressed as a percentage.
Choropleth Map	A map where regions are shaded according to a data value, e.g., case count.
EDA	Exploratory Data Analysis — the process of summarising and visualising a dataset's main characteristics.
DataFrame	A Pandas data structure representing tabular data with labelled rows and columns.
KPI	Key Performance Indicator — a summary metric highlighted for quick reference (e.g., total deaths).